Contribution of Gear Body to Tooth Deflections—A New Bidimensional Analytical Formula

P. Sainsot and P. Velex
Contact and Solid Mechanics Laboratory, UMR CNRS 5514 INSA de Lyon, 20 Avenue Albert Einstein, 69 621 Villeurbanne Cedex, France
e-mail: Philippe.Velex@insa-lyon.fr

O. Duverger
Pôle Machines et Commandes, CETIM Senlis, France

The magnitude and variation of tooth pair compliance affects tooth loading and gear dynamics significantly. This paper presents an improved fillet/foundation compliance analysis based on the theory of Muskhelishvili applied to circular elastic rings. Assuming linear and constant stress variations at root circle, the above theory makes it possible to derive an analytical formula for gear body-induced tooth deflections which can be directly integrated into gear computer codes. The corresponding results are in very good agreement with those from finite element models and the formula is proved to be superior to Weber's widely used equation, especially for large gears. [DOI: 10.1115/1.1758252]

1 Introduction

In 1949, Weber [1], and later Weber and Banascheck [2] proposed a method of calculating tooth deflections of spur gears by superimposing displacements which arise from i) the contact between the teeth, ii) the tooth itself considered as a beam and, iii) the gear body regarded as a semi-infinite elastic plane. Extensions and variants of the methodology were introduced by O'Donnell [3–4] with regard to the foundation effects and by Attia [5], Cornell [6] among others, concerning analytical developments.

Weber analyzed the contact compliance by using the Hertzian 2D theory for cylinders in contact with the datum for displacements taken at the tooth center line, and gave the normal approach between the parts in contact under the form:

$$\delta_c = 4\,\frac{F}{b}\,\frac{1-\nu^2}{\pi E}\left[\ln\frac{2\sqrt{k_1 k_2}}{a} - \frac{\nu}{2(1-\nu)}\right] \tag{1}$$

with F: force on the tooth
b: tooth face width
k_1, k_2: the distances on the pinion, on the gear between the point of contact and the tooth center line (Fig. 1)
$a=\sqrt{8\,F/b\,\rho_1\rho_2/\rho_1+\rho_2\,1-\nu^2/\pi E}$: half width of contact zone (in the profile direction)
E, ν: Young's modulus, Poisson's ratio.
ρ_1, ρ_2: radii of curvature (Fig. 1).

The other widely-used formulas for tooth contact deflection comprise the analytical formula of Lundberg [7], the approximate hertzian approach originally used at Hamilton Standard [6] and the semi-empirical formula developed by Palmgren for rollers [6].

Tooth bending deformations were derived by considering the tooth as a cantilever of variable cross-section and equating the strain energy to the work of the external force. The corresponding Weber equation for bending displacements reads:

$$\delta_b = \frac{F}{b}\,\frac{1}{E}\cos^2\alpha_u\left[10.92\int_0^{u_w}\frac{(u_w-y)^2}{d(y)^3}dy + 3.1(1+0.294\,tg^2\alpha_u)\int_0^{u_w}\frac{1}{d(y)}dy\right] \tag{2}$$

with α_u: pressure angle
u_w and $d(y)$ are defined in Fig. 1

For the contribution of gear body, the tooth is supposed to be rigid and the wheel body is modeled as an elastic half plane on which the normal and tangential forces as well as the bending moment are applied. Assuming a linear distribution of normal stress and a constant shear stress at the tooth root, an estimate of the displacement in the direction of the tooth load is given under the form:

$$\delta_{fw} = \frac{F}{b}\,\frac{1}{E}\cos^2\alpha_u\left[L\left(\frac{u_w}{S_{fw}}\right)^2 + M\left(\frac{u_w}{S_{fw}}\right) + P(1+Q\,tg^2\alpha_u)\right] \tag{3}$$

with S_{fw}: tooth thickness at the critical section according to Weber (Fig. 1)
L, M, P and Q: constants which slightly differ depending on the authors as indicated in Table 1.

Although the equations above are based on a simplified bidimensional approach, they are still widely used in gear design. For example, the mesh stiffness formulas in the ISO standard 6336 [8] stem from (1)–(3), which were modified to bring the values in closer agreement with the experimental results.

The objective of this technical brief is to extend the present theory about the contributions of gear body by developing an original semi-analytical formula which is not based on the half plane hypothesis, but which accounts for actual solid disk wheels. The validity and the practical interest of this formula are illustrated by comparisons with some results from the original Eq. (3) and 2D finite element calculations.

2 Theory

2.1 Solution for Elastic Rings. Muskhelishvili [9] presented a general bidimensional solution for circular rings based on a complex power series representation of displacements, stresses and external loads. The particular case of interest in gearing corresponds to the mixed problem of calculating radial ($u_r(\theta)$) and tangential ($u_t(\theta)$) displacements at the outer boundary of an elastic ring in terms of the normal and tangential stresses, i.e., $\sigma_{rr}(\theta)$ and $\sigma_{rt}(\theta)$ applied to the same boundary at the outer radius R_f (Fig. 2). Assuming that the displacements at the inner radius R_a are nil, one obtains:

$$u_r(\theta)+i\,u_t(\theta)=\frac{Rf}{2\mu}\sum_{k\in Z}U_k e^{ik\theta}=\frac{Rf}{2\mu}\left[U_0+\sum_{k=1}^{N}U_k e^{ik\theta}+U_{-k}e^{-ik\theta}\right] \tag{4}$$

$$\frac{\partial u_r}{R_f\partial\theta}+i\,\frac{\partial u_t}{R_f\partial\theta}=\frac{i}{2\mu}\left[\sum_{k=1}^{N}k(U_k e^{ik\theta}-U_{-k}e^{-ik\theta})\right] \tag{5}$$

$$\sigma_{rr}(\theta)-i\,\sigma_{rt}(\theta)=\sigma_0+\sum_{k=1}^{N}\sigma_k e^{ik\theta}+\sigma_{-k}e^{-ik\theta} \tag{6}$$

where i is the imaginary number ($i^2=-1$) and μ is a Lamé's coefficient.

For every harmonic k, the stress-displacement relations for an elastic ring can be written as:

$$\begin{Bmatrix} U_k \\ \overline{U_{-k}} \end{Bmatrix} = \begin{bmatrix} U_{A,k} & U_{B,k} \\ U_{B,-k} & U_{A,-k} \end{bmatrix}\begin{Bmatrix} \sigma_k \\ \overline{\sigma_{-k}} \end{Bmatrix} \tag{7}$$

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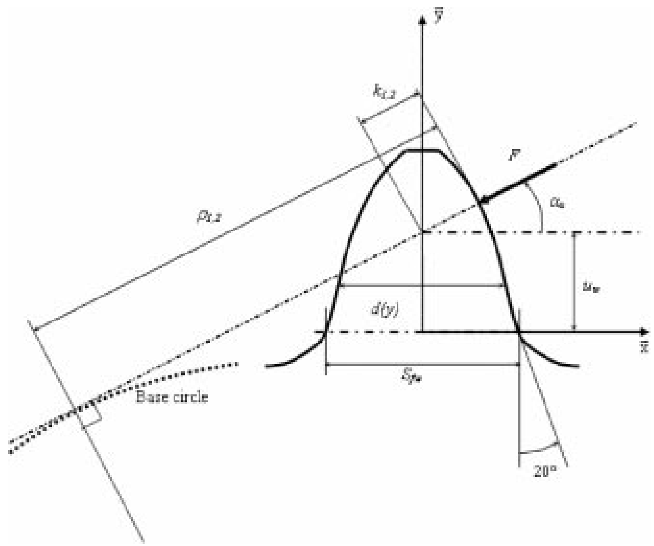

Fig. 1 Geometrical parameters used in Weber-Banaschek's equations

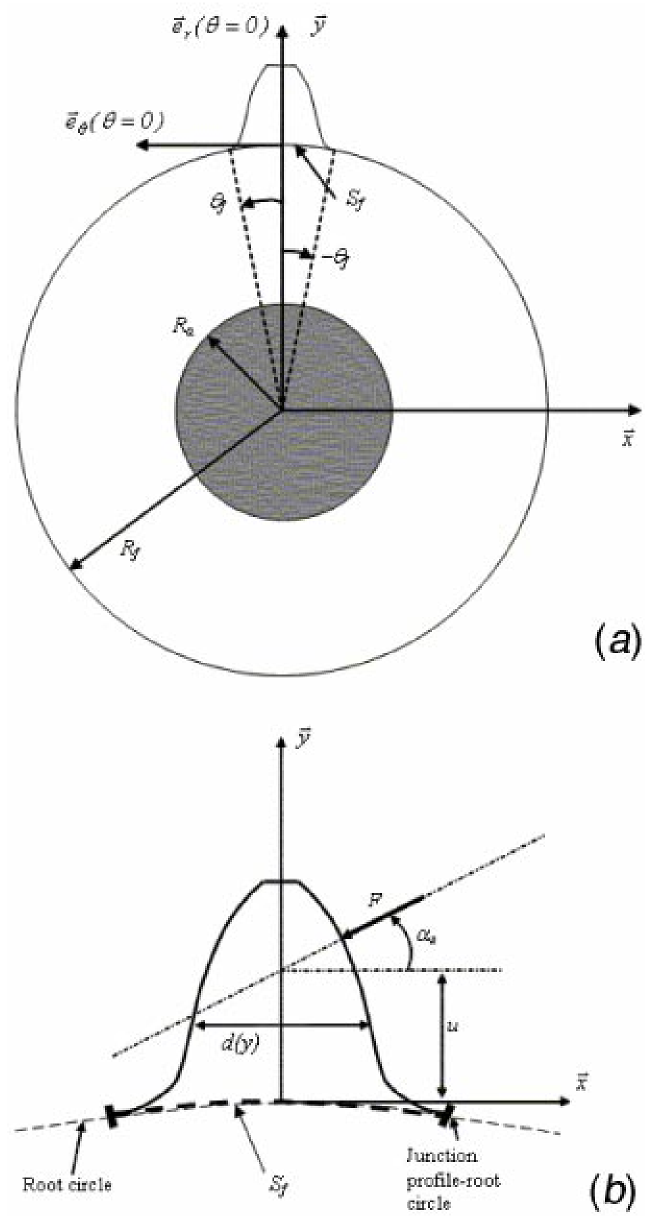

Fig. 2 (*a*) Elastic ring and tooth parameters—*Geometrical parameters for gear body* (*b*) Elastic ring and tooth parameters—*Geometrical parameters for one tooth*

$$\text{with} \qquad U_{A,k}=\frac{(1+\chi)(h^2-1)h^2}{(1-k^2)(h^2-1)^2-(\chi+h^{2+2k})(\chi+h^{2-2k})}$$

$$U_{B,k}=-\frac{\dfrac{\chi(h^{2+2k}-1)(h^{2-2k}+\chi)}{(1+k)}+(1-k)(h^2-1)^2}{(1-k^2)(h^2-1)^2-(\chi+h^{2+2k})(\chi+h^{2-2k})}$$

$$\chi=3-4\nu \text{ for plane strain}$$

$$\chi=\frac{3-\nu}{1+\nu} \text{ for plane stress}$$

$$h=R_f/R_a$$

$\overline{U}_{-k}$ and $\overline{\sigma}_{-k}$ are the conjugates of U_{-k} and σ_{-k} respectively

2.2 External Load and Stress Coefficients. Equations (4) to (7) make it possible to build the displacement field knowing the power series representation of the external loading at the boundary $r=R_f$, i.e., the various components σ_k and $\overline{\sigma}_{-k}$. The influence of a loaded tooth on gear body is taken into account by introducing distributions of normal and tangential stresses over the arc$\lfloor -\theta_f, +\theta_f \rfloor$ along the tooth root as shown in Fig. 2. It is to be stressed that the parameters S_f and u in the proposed formulation (Fig. 2(*b*)) are defined using the root circle and, consequently they differ slightly from the parameters S_{fw} and u_w (Fig. 1) in Weber's equations.

From the force and moment equilibrium of one tooth, the following expressions of the stresses can be derived:

$$\sigma_{rr}(\theta)=-\frac{F}{bS_f}\cos\alpha_u\left[tg\,\alpha_u\varphi_1(\theta)+\frac{u}{S_f}\psi_1(\theta)\right] \qquad (8)$$

Table 1 Constants in Eq. (3) for $\nu=0.3$

	L	M	P	Q
Weber [1] –Attia [5]	5.2	1	1.4	0.294–0.32
Cornell* [6]	5.306	1.4 (plane stress) 1.14 (plane strain)	1.534	0.32

*E for plane stress, $E/(1-\nu^2)$ for plane strain

$$\sigma_{rt}(\theta)=\frac{F}{bS_f}\cos\alpha_u\varphi_2(\theta) \qquad (9)$$

with $\varphi_{1,2}(\theta)$, two even functions and $\psi_1(\theta)$, an odd function of the angular variable θ which are shape functions depending on the assumed stress distributions on S_f and such that $\varphi_{1,2}(\theta)=\psi_1(\theta)=0$ when $\theta\notin\lfloor -\theta_f, +\theta_f\rfloor$.

The decomposition of Eqs. (8)–(9) in Fourier series of period 2π leads to the following coefficients:

$$\sigma_0=\frac{F}{bS_f}\frac{\theta_f}{\pi}\cos\alpha_u\{-tg\,\alpha_u-i\}$$

$$\sigma_k=\frac{F}{bS_f}\frac{\theta_f}{\pi}\cos\alpha_u\left\{-\Lambda_{k,\theta_f}tg\,\alpha_u-i\left[\Gamma_{k,\theta_f}-\frac{u}{S_f}\Delta_{k,\theta_f}\right]\right\} \qquad (10)$$

$$\sigma_{-k}=\frac{F}{bS_f}\frac{\theta_f}{\pi}\cos\alpha_u\left\{-\Lambda_{k,\theta_f}tg\,\alpha_u-i\left[\frac{u}{S_f}\Delta_{k,\theta_f}+\Gamma_{k,\theta_f}\right]\right\}$$

where $$\Lambda_{k,\theta_f}=\frac{1}{2\theta_f}\int_{-\theta_f}^{+\theta_f}\varphi_1(\theta)\cos(k\theta)d\theta$$

$$\Delta_{k,\theta_f}=\frac{1}{2\theta_f}\int_{-\theta_f}^{+\theta_f}\psi_1(\theta)\sin(k\theta)d\theta$$

$$\Gamma_{k,\theta_f}=\frac{1}{2\theta_f}\int_{-\theta_f}^{+\theta_f}\varphi_2(\theta)\cos(k\theta)d\theta$$

2.3 Displacement in the Tooth Force Direction. Following the usual decomposition into tooth bending and gear body-induced deflections, the contribution of gear body to the tooth displacement at the intersection of the tooth center line and the force line of action, projected in the direction of the tooth force, can be expressed as:

$$\delta_f=\left[\left(1+\frac{u}{R_f}\right)u_t(\theta=0)-\frac{u}{R_f}\frac{\partial u_r(\theta=0)}{\partial\theta}\right]\cos\alpha_u - u_r(\theta=0)\sin\alpha_u \tag{11}$$

which depends on the displacements and their derivatives at $\theta=0$. In such conditions, the expressions (4) and (5) are largely simplified, and after some manipulations, one obtains:

$$u_r(\theta=0)=-\frac{1}{2\mu}\frac{F}{b}\frac{\cos\alpha_u}{2\pi}tg\alpha_u\left[\frac{(\chi-1)(h^2-1)}{(\chi-1+2h^2)}+\sum_{k=1}^{N}[\lambda_{k,h}\Lambda_{k,\theta_f}]\right] \tag{12}$$

$$u_t(\theta=0)=\frac{1}{2\mu}\frac{F}{b}\frac{\cos\alpha_u}{2\pi}\left[(h^2-1)+\sum_{k=1}^{N}\left[\frac{u}{S_f}\delta_{k,h}\Delta_{k,\theta_f}-\gamma_{k,h}\Gamma_{k,\theta_f}\right]\right] \tag{13}$$

$$\frac{\partial u_r(\theta=0)}{\partial\theta}=-\frac{1}{2\mu}\frac{F}{b}\frac{\cos\alpha_u}{2\pi}\left[\sum_{k=1}^{N}k\left[\frac{u}{S_f}\lambda_{k,h}\Delta_{k,\theta_f}+\delta_{k,h}\Gamma_{k,\theta_f}\right]\right] \tag{14}$$

with:

$$\lambda_{k,h}=\lfloor U_{A,k}+U_{A,-k}+U_{B,k}+U_{B,-k}\rfloor$$

$$\delta_{k,h}=[U_{A,k}-U_{A,-k}+U_{B,k}-U_{B,-k}]$$

$$\gamma_{k,h}=[U_{A,k}+U_{A,-k}-U_{B,k}-U_{B,-k}]$$

3 Analytical Formula

The expressions of Λ_{k,θ_f}, Δ_{k,θ_f} and Γ_{k,θ_f} depend on the assumed distributions of normal and tangential stresses on S_f which are unknown. This point was discussed by O'Donnell [3–4], Seager [10], both of whom concluded that cubic variations of normal stress and parabolic distributions of shear stress along S_{fw} were more realistic than the linear, respectively constant distributions considered by Weber. However, in order to establish an analytical formula, a linear variation of normal stress and a constant shear stress along S_f on the root circle are considered. This leads to:

$$\Lambda_{k,\theta_f}=\Gamma_{k,\theta_f}=\left[\frac{\sin(k\theta_f)}{k\theta_f}\right] \tag{15}$$

$$\Delta_{k,\theta_f}=\frac{6\lfloor\sin(k\theta_f)-k\theta_f\cos(k\theta_f)\rfloor}{k^2\theta_f^2} \tag{16}$$

and, for a given material, the displacement δ_f can be expressed under the same form as (3) but the constants L, M, P and Q are

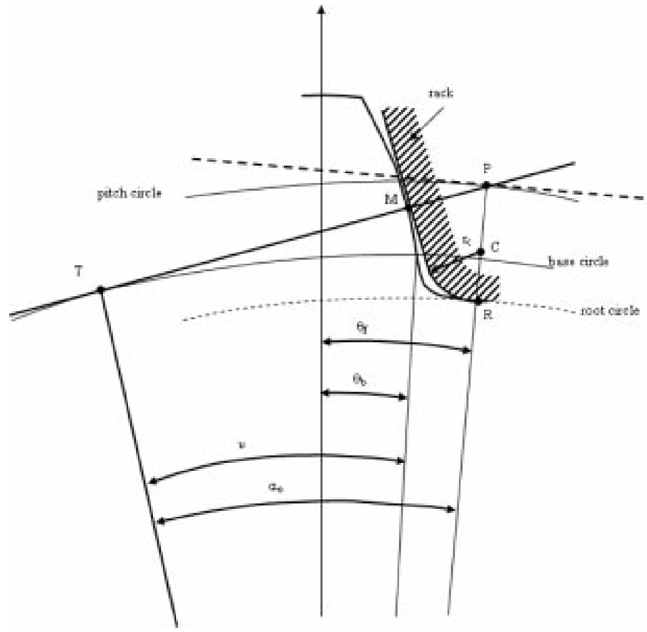

Fig. 3 Geometric parameters needed for the expression of θ_f

replaced by 4 functions $L^*(h,\theta_f)$, $M^*(h,\theta_f)$, $P^*(h,\theta_f)$ and $Q^*(h,\theta_f)$ depending on $h=R_f/R_a$ and θ_f only. It follows that:

$$\delta_f=\frac{1}{E}\frac{F}{b}\cos^2\alpha_u\left[L^*(h,\theta_f)\left(\frac{u}{S_f}\right)^2+M^*(h,\theta_f)\frac{u}{S_f}+P^*(h,\theta_f)\times[1+Q^*(h,\theta_f)tg^2\alpha_u]\right] \tag{17}$$

with $$L^*(h,\theta_f)=\frac{1+\nu}{\pi}\theta_f\left[\sum_{k=1}^{N}k\lambda_{k,h}\Delta_{k,\theta_f}+\sum_{k=1}^{N}\delta_{k,h}\Delta_{k,\theta_f}\right]$$

$$M^*(h,\theta_f)=\frac{1+\nu}{\pi}\theta_f\left[h^2-1-\sum_{k=1}^{N}(\gamma_{k,h}\Gamma_{k,\theta_f}-k\delta_{k,h}\Gamma_{k,\theta_f})+\frac{1}{2\theta_f}\sum_{k=1}^{N}\delta_{k,h}\Delta_{k,\theta_f}\right]$$

$$P^*(h,\theta_f)=\frac{1+\nu}{2\pi}\left[h^2-1-\sum_{k=1}^{N}\gamma_{k,h}\Gamma_{k,\theta_f}\right]$$

$$Q^*(h,\theta_f)=\left[\frac{(\chi-1)(h^2-1)}{\chi-1+2h^2}+\sum_{k=1}^{N}\lambda_{k,h}\Lambda_{k,\theta_f}\right]\Big/\left[h^2-1-\sum_{k=1}^{N}\gamma_{k,h}\Gamma_{k,\theta_f}\right]$$

Referring to Fig. 3, the angle θ_f between the tooth center-line and the junction with the root circle is given by:

$$\theta_f=\alpha_0-\nu+\theta_b \tag{18-a}$$

with: $\theta_b=1/2\,s_b/r_b=1/Z[\pi/2+2xtg\alpha_0]+inv\,\alpha_0$, where s_b is the base thickness, r_b is the base radius, x is the profile shift coefficient and α_0 is the rack pressure angle

$\nu=\dfrac{TM}{r_b}=\dfrac{2TM}{m_nZ\cos\alpha_0}$, where m_n is the module and Z is the number of teeth

The expression of TM is deduced of:

Table 2 Coefficients of the polynomial curve fitting

	A_i	B_i	C_i	D_i	E_i	F_i
$L^*(h,\theta_f)$	$-5.574E-5$	$-1.9986E-3$	$-2.3015E-4$	$4.7702E-3$	0.0271	6.8045
$M^*(h,\theta_f)$	$60.111E-5$	$28.100E-3$	$-83.431E-4$	$-9.9256E-3$	0.1624	0.9086
$P^*(h,\theta_f)$	$-50.952E-5$	$185.50E-3$	$0.0538E-4$	$53.300E-3$	0.2895	0.9236
$Q^*(h,\theta_f)$	$-6.2042E-5$	$9.0889E-3$	$-4.0964E-4$	$7.8297E-3$	-0.1472	0.6904

$$TM=TP-PM \tag{18-b}$$

with: $TP=m_n Z/2 \sin\alpha_0$
$PM=m_n\overline{r_c}+PC\sin\alpha_0$
$PC=m_n(h_a-x-\overline{r_c})$
h_a is the tool addendum coefficient and $\overline{r_c}=r_c/m_n$ is the dimensionless tool tip radius

The combination of (18-*a*) and (18-*b*) finally leads to:

$$\theta_f=\frac{1}{Z}\left[\frac{\pi}{2}+2tg\,\alpha_0(h_a-\overline{r_c})+\frac{2\overline{r_c}}{\cos\alpha_0}\right] \tag{18-c}$$

which indicates that, apart from the radius ratio h, the functions L^*, M^*, P^* and Q^* depend mainly on the number of teeth Z, and to a lesser extent to the tool radius $\overline{r_c}$, the pressure angle α_0 and the tool addendum h_a which, in practice, have limited ranges of variation.

For plane strain conditions, $L^*(h,\theta_f)$, $M^*(h,\theta_f)$, $P^*(h,\theta_f)$ and $Q^*(h,\theta_f)$ have been numerically calculated over a realistic range of values (h between 1.4 and 7, θ_f between 0.01 and 0.12) and curve-fitted by polynomial functions of the form:

$$X_i(h,\theta_f)=A_i/\theta_f^2+B_i h^2+C_i h/\theta_f+D_i/\theta_f+E_i h+F_i \tag{19}$$

with the constants A_i, B_i, C_i, D_i, E_i and F_i being given in Table 2.

The validity of Eqs. (17)–(19) for calculating gear-body induced tooth deflections is demonstrated by comparing the analytical results with those from finite element models (Fig. 4) for a wide range of geometrical parameters. For each geometry, a force per unit of face width $F/b=1E6N/m$ is applied at three positions on the profile which approximately correspond to the tooth tip, the pitch point and the active point the nearest to the tooth root. As observed in Tables 3, 4 and 5, very good agreements are found with maximum relative deviations generally below 10% with the exception of a force on tooth tip for $Z=26$ and $h=2$. Weber's equation (3) generally leads to lower displacements especially when h becomes large.

The bounds of variation of the functions $L^*(h,\theta_f)$, $M^*(h,\theta_f)$, $P^*(h,\theta_f)$ and $Q^*(h,\theta_f)$ are:

$$6.82\leq L^*(h,\theta_f)\leq 6.94$$
$$1.08\leq M^*(h,\theta_f)\leq 3.29$$
$$2.56\leq P^*(h,\theta_f)\leq 13.47$$
$$0.141\leq Q^*(h,\theta_f)\leq 0.62 \tag{20}$$

Compared with the values in Table 1, similar orders of magnitude are found for $L^*(h,\theta_f)$, $M^*(h,\theta_f)$ and $Q^*(h,\theta_f)$ but $P^*(h,\theta_f)$, which mostly accounts for gear body torsional displacements, is largely underestimated by the theory of Weber and the like, especially for the largest amplitudes of h.

4 Conclusion

Based on Muskhelishvili's theory applied to elastic rings, a new analytical bidimensional formula has been set up which allows fast and accurate calculations of gear body-induced tooth deflections. Although the theory relies on rather strong approximations concerning the stress distributions at the root circle, the results are in very good agreement with those given by finite element models. It is also shown that the proposed formula yields better results than Weber's equation based on the semi-infinite plate approximation, particularly for large wheels. The formula can be readily implemented in gear computer codes and be used for determining

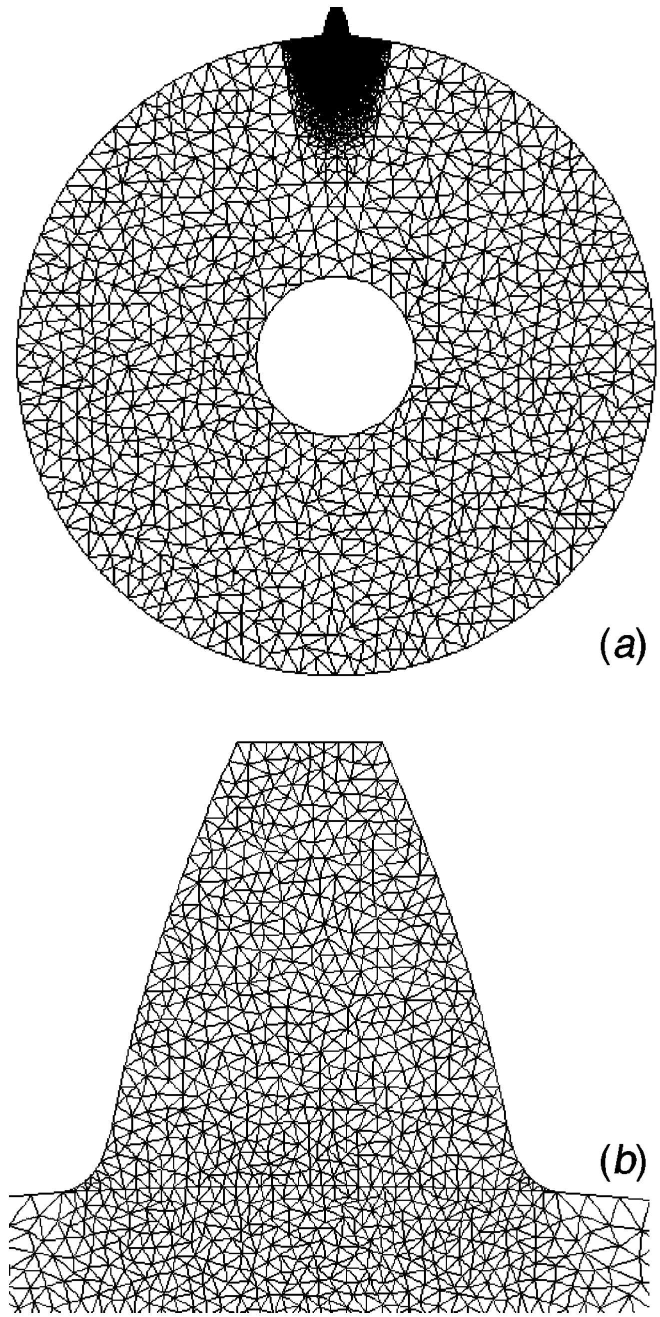

Fig. 4 (*a*) Example of finite element model ($Z=57, h=4, \bar{r}_c=0.2$)—*Global model* (*b*) Example of finite element model ($Z=57, h=4, \bar{r}_c=0.2$)—*Detail on one tooth*

Table 3 Comparisons with Weber's formula and finite element results ($\alpha_0=20°$, $h_a=1.4$, $\bar{r}_c=0.2$, $m_n=1$ mm, $F/b=1E6$ N/m)

Z (θ_f) in rad	h	Radius at the point of application of the force (mm)	δ_{fw} (μm) (3)	δ_f FEM (μm)	δ_f Analytical (μm)	$\frac{\lvert\delta_f\text{FEM}-\delta_f\rvert}{\delta_f\text{FEM}}$ (%)
26	2	12.24	8.45	15.78	16.02	1.5
(0.11)		13.03	13.31	21.95	20.74	0.6
		13.91	20.55	32.52	27.1	16.7
157	7	77.53	6.4	59.64	60.02	0.9
(0.018)						
		78.51	11.19	65.83	65.46	0.6
		79.39	19.5	77.21	73.79	4.4

Table 4 Comparisons with Weber's formula and finite element results. Influence of h ($\theta_f=0.0503$ with $Z=57$, $\alpha_0=20°$, $h_a=1.4$, $\bar{r}_c=0.2$, $m_n=1$ mm, $F/b=1E6$ N/m)

h	Radius at the point of application of the force (mm)	δ_{fw} (μm) (3)	δ_f FEM (μm)	δ_f Analytical (μm)	$\frac{\lvert\delta_f\text{FEM}-\delta_f\rvert}{\delta_f\text{FEM}}$ (%)
2	27.55	6.76	14.75	15.56	5.5
	28.48	11.54	20.49	20.86	1.8
	29.4	19.79	31.84	28.9	9.2
4	27.56	6.76	28.91	28.55	1.2
	28.48	11.54	35.13	33.65	4.2
	29.4	19.8	46.68	41.7	10.7

Table 5 Comparisons with Weber's formula and finite element results. Influence of the normalized tool tip radius (with $Z=57$, $\alpha_0=20°$, $h_a=1.4$, $h=2$, $m_n=1$ mm, $F/b=1E6$ N/m)

$\bar{r}_c$ (θ_f) in rad	Radius at the point of application of the force (mm)	δ_{fw} (μm) (3)	δ_f FEM (μm)	δ_f Analytical (μm)	$\frac{\lvert\delta_f\text{FEM}-\delta_f\rvert}{\delta_f\text{FEM}}$ (%)
0 (0.0454)	28.5	12.57	21.39	22.78	6.5
0.2 (0.0503)	28.48	11.54	20.49	20.86	1.8
0.4 (0.0553)	28.52	11.09	20.61	19.86	3.6

mesh stiffness, load distributions on tooth flanks and transmission errors under load. In order to assess the influence of the assumed stress distributions at the root circle, cubic normal stress and parabolic shear stress distributions were considered but, compared with the finite element solutions of reference, very limited improvements on δ_f were found and no simple deflection formula can be obtained. In such conditions, the proposed analytical approach has no real advantage over the bidimensional solutions by finite element models which are more general and flexible, and can directly account for tooth and body-induced deflections.

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